Validation and AI Use — Joby Aviation (JOBY), FIN 43900 Project 1

Tobey Uhm · This Markdown file is the working source; Brightspace `Validation-and-AI-Use.pdf` is the frozen record of the same content.

1. Decision, version, and as-of boundary

Field	Value
Decision / user	Initiate, watch/defer, or do not initiate JOBY, for a buy-side investment committee with no current position
Result	**Watch / defer**; $0.00–$3.82 per share (base $0.34) vs $6.69
Valuation date / price	Sept. 1, 2026 / $6.69 close (S9)
Source cutoff	Latest filing on or before Sept. 1, 2026: Q2 2026 10-Q, filed Aug. 6, 2026 (balance sheet June 30, 2026; cover shares Aug. 3, 2026). Nothing after Sept. 1, 2026 is used.
Repository / commit	`Joby-research`, branch `main`; frozen submission = git tag **`project1-submission`** (hash listed in the manifest)
Run	`python analysis.py` → `output/visible_output.json`, `output/joby_valuation_report.md` (timestamp printed in the report)
Environment	Python 3.13.15 (Windows 11), standard library only for the engine; Streamlit ≥1.40,<2 for the app

2. Data and convention ledger

Material input	Source	As-of / availability	Definition / units	Transformation / exclusion	Risk
Cash + short-term investments $2,263.826M	S4 10-Q balance sheet p. 3	June 30, 2026; public Aug. 6	USD M	Restricted cash ($1.3M current, $4.9M non-current) excluded. All counted as non-operating because H2 2026 burn is already in FCFF.	Liquidity is runway, not surplus. Shown via the funding schedule.
2032 convertible notes $690.0M principal, 0.75%, conversion price $14.19, 48.63M if-converted shares; fair value $628.0M	S4 Note 6	June 30, 2026	USD M; shares M	Debt at principal in the bridge (value < $14.19). Fair value used for the WACC weight.	Treatment flips only if value > $14.19 (coded and tested).
Mortgage $30.75M at 6.784%	S4 Note 6	June 30, 2026	USD M	Debt	—
Warrants $21.505M; Blade EBITDA earnout up to $17.5M	S4 balance sheet, Notes 4 and 7	June 30, 2026	USD M	Warrants deducted at fair value, not added as shares. Earnout deducted at maximum. Sponsor earnout shares are already in the basic count, so their liability is not deducted.	Double counting is avoided by design.
Shares: basic 989.124M; RSUs 55.886M; RSAs 0.446M; early-exercised 0.533M; options 5.270M	S4 cover page and antidilution table	Aug. 3 / June 30, 2026	M shares	Options counted at full dilution (strike not extracted)	Overstates dilution by < 0.5%.
H1 2026 actuals: revenue 62.885, COGS 47.101, R&D 372.132, SG&A 138.115, D&A 22.853, capex 106.559	S4 statements pp. 4, 8	Six months to June 30, 2026	USD M	Removed from the FY2026 plan to form the July–Dec stub	—
Federal NOL $1,662.0M	S3 FY2025 10-K tax note	Dec. 31, 2025	USD M	H1 2026 loss added; 80% usage limit	State NOLs ignored (conservative)
Risk-free 4.20%	S4 Note 3 (warrant valuation input)	June 30, 2026	Annual rate	Used as rf in CAPM	Market rate on Sept. 1 may differ by a few bp
Beta 1.8; ERP 5.0%; Kd 8%; tax 24%	J1 (judgment)	Sept. 1, 2026	Decimals	Tornado tests β 1.4–2.2 and ERP 4–6%	Discount rate is the second-largest value driver
FY2026 services revenue $120M	S7 guidance $115–125M (midpoint)	Aug. 12, 2026	USD M	—	Management claim
Fleet, hours, $/flight hour, margins, opex, capex 2026–31; convergence; terminal	J2, J3 (judgment)	Sept. 1, 2026	See inputs JSON	Fleet anchored to "5 aircraft flying, 12 in production" (S7)	Largest value driver. Reverse DCF quantifies it.
Price $6.69	S9 (recorded in Lab 08, commit	Sept. 1, 2026 close	USD	—	Single-day price

Material input	Source	As-of / availability	Definition / units	Transformation / exclusion	Risk
	a6e9520)				

Date discrepancy found and corrected: Lab 06 and the Lab 11 code cite the FY2025 10-K as "filed Feb. 26, 2026". EDGAR's filing index shows **2026-02-27** (Edition A was right). Edition B and the source ledger use Feb. 27.

3. Validation register

Claim / output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
Discounting and Gordon-terminal convention	Terminal value discounted 6 years; wrong rate	Week 3 known answer: 12 lines within $0.01; $27.4974; WACC 11% → $23.41; reverse shift ≈ +1.78 pts	All match ($27.4974; +1.7779 pts)	Pass	`tests/test_model.py::KnownAnswer`
Enterprise-to-equity sign convention	Cash subtracted instead of added	Edition A synthetic bridge = $8.20; Joby bridge cash > 0 and rebuilds to equity value	$8.20; rebuild exact	Pass	`test_synthetic_bridge_from_edition_a`, `Bridge` tests
Lab 11 defect 1	`zip()` silently drops a fifth year	Four-value path must raise	Raises in the new model and the repaired legacy script	Fixed	`test_defect1_four_year_path_rejected`
Lab 11 defect 2	Terminal value discounted at the module constant, explicit years at the input rate	With r = 20%, PV equals a hand formula using one rate	Equal to 6 dp	Fixed	`test_defect2_single_discount_rate`
Lab 11 defect 3	g ≥ r accepted → negative or infinite terminal value	g = 13% vs r = 13% must raise; g > 4% rejected	Raises	Fixed	`test_defect3_…`, `test_growth_above_gdp_rejected`
Repair does not change the Lab 11 record	Fix alters published results	2030 operating profit $375.1M; FCFE $121.2M	Unchanged	Pass	`test_legacy_lab11_results_unchanged_by_repair`; `legacy/repaired_run_output.txt`
Linked statements	Balance sheet does not balance; cash below floor	|gap| < $0.01M every period, every scenario; cash ≥ $500M floor	Max gap 0.0000	Pass	`Accounting` tests; statement table in the report
Stub construction	H1 double counted	H1 opex reconciles to the 10-Q operating loss −$494.463M;	Exact	Pass	`test_h1_actuals_reconcile…`, `test_stub_plus_h1…`

Claim / output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
		stub + H1 = FY2026 plan			
Scenario independence	A scenario mutates the base inputs	Deep-copy check	Base unchanged	Pass	`test_inputs_not_mutated_by_scenarios`
Economic direction	Wrong-sign driver	Higher WACC ↓ value; delay ↓; acceleration ↑; more fleet ↑	All as expected	Pass	`Directional` tests
FY2026 cash use vs guidance	Model burn inconsistent with management	H2 2026 modeled cash use (ex-SBC, incl. interest income) vs $385–415M guided	First draft (capex $230M) about $90M heavier than guidance. FY2026 capex reset to $200M (H2 $93M). Remaining gap ≈ SBC treated as cost.	Qualified	`data/joby_inputs_2026-09-01.json` capex
Convergence design (highest-risk modeling claim)	Linear ratio interpolation ignores fixed-cost leverage	Review whether opex scales like a fixed cost after certification	Draft gave −$1.21/share with R&D at 33% of revenue in 2031 falling only linearly. Redesigned: R&D +2%/yr and SG&A at half of revenue growth, with 8%/10% floors → $0.34. The change rests on cost structure, not on matching the price (still 5% of price).	Corrected	Draft was not committed; the redesign is in the first commit `8fc1a0a` (`joby/model.py`, convergence block)
Highest-risk decision claim: "price requires more scale than evidence supports"	Result driven by WACC, not scale	Reverse DCF on fleet with WACC held; WACC × g grid	2.13× fleet needed; no grid cell (10.7–14.7%, 2–4%) reaches $6.69	Supports claim	Report: Reverse DCF, grid
Recommendation-changing downside	Downside is cosmetic	Two-year certification delay through the linked statements	EV −$7.39B; new equity $12.7B; action → **Do not initiate**	Recommendation changes	Report: Scenarios
Reproducibility (cold run)	A clean user following the README cannot reproduce the result	Fresh copy of the repo in a new folder, new virtual environment, then install requirements → run `python analysis.py` → run the tests → start the app. Expected: all four steps work.	**First failure:** `python analysis.py` crashed while printing the report. The Windows console (cp1252) could not print the "−" minus sign (`UnicodeEncodeError`). **Fix:** print as UTF-8 (commit `8f813c3`). **Rerun:** analysis OK, 23 of 23 tests pass, app starts (HTTP 200). About 3 minutes including install. The same crash later hit `locked_test.py` and got the same fix (commit `ab64e4d`).	Fixed	Section 3a below (cold-run log)
App rendering	$...$ rendered as	Visual check of the range	"$0.00–$3.82" was shown as math. Dollar	Fixed	`app.py`

Claim / output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
	LaTeX math in Streamlit	metric	signs escaped (commit after `8f813c3`).		

3a. Cold-run log (Oct. 19, 2026, ET)

```
COLD RUN START 2026-10-19T20:12:40-04:00   (fresh clone into a new folder; new venv)
pip OK
UnicodeEncodeError: 'charmap' codec can't encode character '−'   <- first failure (analysis.py print)
RERUN after fix 2026-10-19T20:16:00-04:00 commit 8f813c3
analysis OK
Ran 23 tests ... OK
streamlit health HTTP 200
```

Limitation: this cold run used a fresh clone and environment on the author's machine. A README-only run by another person was not recorded.

4. Locked Changed-Input Record — human prediction first; AI closed during execution

4A. Final-model record (Project 1 model)

The test runner (`locked_test.py`) fixes the material input before anyone predicts: **revenue per eVTOL flight hour, all years, $2,500 → $2,000 in 2027 (×0.8), same escalation**. Nothing else changes. This input is deliberately absent from every sensitivity table in the frozen output, so the result was not visible before the prediction. The runner refuses to execute until `docs/locked-prediction.md` has no blanks and is committed. It records that commit hash and writes `output/locked_test_result.md`.

Precommit before the run and before any AI assistance on the prediction

Prediction written Oct. 19, 2026, 9:21 PM ET by Tobey Uhm; AI assistance on the prediction: none. Full text: `docs/locked-prediction.md`.

Precommit timestamp	Repository commit	Material input	Old → new value / units	Expected output direction	Expected decision effect
2026-10-19 21:22:24 ET (`1796d19`). A follow-up commit at 21:22:31 (`a0f877b`) removed placeholder text from the instruction line only; the prediction is unchanged. The runner recorded `a0f877b`.	`1796d195b752...`, `a0f877b5131a...`	Revenue per eVTOL flight hour	$2,500 → $2,000 USD per flight hour in 2027 (×0.8); +2.5%/yr thereafter	Value per share **down**, "significantly below $0.34 ... likely deeply negative or toward $0.00". Cumulative new equity **up**. Mechanism: costs unchanged, so gross profit absorbs the entire revenue loss.	**Unchanged: Watch / defer**

Preserved execution result

Run 2026-10-19 21:23:20 ET, commit `ab64e4d`; result committed in `2227cf7`. A first run at 21:23:03 ET wrote the same result files and then crashed while printing to the console (the same Windows cp1252 `UnicodeEncodeError` found in the cold run). The print was fixed in `ab64e4d` and rerun. Before/after outputs were checked identical to the first run.

Output	Before	After	Change
Value per share (unfloored)	$0.34	**−$1.34** (floored $0.00)	−$1.67
Enterprise value ($M)	−1,151.7	−2,911.3	−1,759.6
2030 revenue ($M)	1,132.5	938.7	−193.9
2030 FCFF ($M)	−1,020.2	−1,091.7	−71.5
Cumulative new equity ($M)	5,028.1	5,937.4	+909.3
Committee action	Watch / defer	**Do not initiate**	**changed**

Prediction reconciliation	Failure diagnosis	Evidence path
Value direction: correct. Value fell from $0.34 to −$1.34; modeled equity is below zero, so the shares are worth about $0. The predicted size ("deeply negative or toward $0.00") was right. **New equity: correct.** It rose by $909M. **Mechanism: partly wrong.** I predicted that unchanged	**Why value still fell so much** despite the smaller per-year hit: the loss compounds. Revenue is 18–20% lower every year, and the fixed R&D/SG&A costs stay the same, so operating leverage is lost and the convergence stage starts from a smaller 2031 base. PV of explicit FCFF fell about $1.09B and PV of terminal value fell about $0.80B. **Why my decision prediction failed:** I reasoned that worse economics "reinforce" deferral,	`output/locked_test_result.md`, `output/locked_test_result.json`; prediction `docs/locked-prediction.md`; commits `1796d19`, `a0f877b`, `ab64e4d`, `2227cf7`

Prediction reconciliation	Failure diagnosis	Evidence path
costs would make gross profit absorb 100% of the lost revenue. In the model, eVTOL cost of revenue is set as revenue × (1 − gross margin), so costs fall with price. In 2030 gross profit fell $77.5M on a $193.9M revenue drop (40% = the 2030 margin), not $193.9M. This is the case my item 5 said I would check first. **Decision effect: wrong.** I predicted Watch / defer would hold; it changed to **Do not initiate**.	but I did not apply the committee rule: an action of "Do not initiate" applies whenever value is ≤ $0. The base equity cushion was only $352M, so a $1.76B fall in EV wipes it out. **Model limitation exposed:** holding margin % constant when price falls is generous. Pilot, maintenance and energy cost per flight hour do not fall with fares, so the true downside is likely **larger** than modeled. The next model version should split eVTOL cost into a per-flight-hour cost plus a margin. **Decision consequence:** revenue per flight hour is decision-critical. Value reaches $0 at about **×0.956 (~$2,390 per flight hour in 2027)**, only 4.4% below the base assumption. This adds a monitoring trigger: if disclosed early revenue per flight hour is below about $2,390, move to **Do not initiate**.	

Post-run analysis (the cost trace by year and the breakeven price search) was done with Claude Opus 5.5 after execution and checked against the result files. The prediction itself had no AI assistance.

4B. Lab 11 record (supporting evidence, preserved unchanged)

On the Lab 11 FCFE model: commercialization revenue −30% each year ($75/300/800/1,800/3,500M → $52.5/210/560/1,260/2,450M). **Prediction:** 2030 operating profit and FCFE both fall by about $577.5M; equity funding rises. **Actual:** operating profit −$577.5M (as predicted). FCFE −$498.7M, because the lower case pays no 21% cash tax. **Reconciliation:** direction right; FCFE size overstated by the $78.8M tax effect. Full record: `04_Completed_Joby_work/joby_locked_changed_input_record.md`. Recorded timestamp 2026-09-29T13:56:53-04:00.

Partner: Eric Boyd (esboyd@purdue.edu), NuScale partner, reviewed the prediction and ranges **before execution** and the result afterward. On Oct. 19, 2026 the student confirmed that the quoted Lab 11/12 exchanges accurately reflect what occurred.

Chronology disclosure: the earliest preserved commit (`c8d96ec`, 2026-09-29 14:05:22) already contains both the prediction and the results. The pre-run state is therefore attested by the student and partner, not by an independent pre-execution snapshot. Record 4A exists to provide a properly committed prediction.

5. Named AI-use register

Tool / model surface and exposed version	Interaction date	Material task	Output / claim used or considered	Independent check	Disposition	Effect on decision
OpenAI Codex — **[student: exact surface/model shown]**	Sept. 3, 2026 (Lab 04 report); Sept. 10–29 (Labs 06–11 code)	Drafted the Lab 04 report; generated `dcf.py`, `peer_pe.py`, `joby_proforma_2026.py`	Report prose; Lab 11 pro forma and sensitivity code	Sources checked by student (report footer); Week 3 known answers; partner recomputation in Lab 11	Accepted with checks. Three defects later found (below).	Built the Lab 11 FCFE model that Edition B replaces
ChatGPT — **[student: model/version shown]**	**[student: date]**, Oct. 2026	Reviewed the evidence package; identified gaps and three code defects; drafted partner-exchange wording for confirmation	"Defects: four-year path accepted; inconsistent discount rate; g ≥ r accepted." "FCFE model ≠ required FCFF/WACC."	Each defect reproduced by a failing test, then repaired; student confirmed the partner quotes on Oct. 19	Defects **accepted** (verified); the FCFF requirement **accepted**	Triggered the FCFF rebuild
Anthropic Claude, Claude Code desktop app, model **Claude Opus 5.5 (`claude-opus-5-5`)**	Oct. 19, 2026	Extracted Q2 2026 10-Q and FY2025 10-K data; designed and coded the FCFF/WACC model, tests, app and locked-test runner; drafted README, memo, Edition B, this record and the video scripts	(a) 10-Q figures: cash + ST investments $2,263.826M, 989,124,237 shares, $690M notes at $14.19, antidilution table. (b) A convergence design giving −$1.21/share. (c) A first FY2026 capex of $230M.	(a) The AI's web summary was re-checked against the downloaded SEC HTML text; every bridge number was matched to the filing line. (b) Reviewed against cost behavior. (c) Compared with H2 cash-use guidance. (d) 23 tests plus a cold run.	(a) **Accepted** after verification. (b) **Corrected** (fixed-cost leverage). (c) **Corrected** to $200M.	Produced the valuation range and decision evidence. The student reviewed and owns all assumptions and the recommendation.

No credentials, private data or full chat logs are included. Video scripts drafted with Claude were read and adapted by the student, who owns every spoken claim.

6. Limitations, monitoring, and kill / escalation rules

- **Judgment-heavy ramp:** fleet, utilization, revenue per flight hour and margins are not disclosed. Replace them with reported figures after two paid quarters.
- **Not modeled:** option strikes (full dilution assumed); the Resonant acquisition (NPV-zero); Toyota's pending $250M (excluded); state NOLs; lease liabilities (treated as operating); the 2-month gap between the June 30 balance sheet and the Sept. 1 valuation date.
- **Discount rate:** β and ERP are judgment. ±0.4 β moves value by about $1–1.3/share.
- **Kill / escalation: Do not initiate** if there is no FAA type certificate by Dec. 31, 2027, or if liquidity falls below 12 months of cash use without committed funding. **Escalate for an initiate review** only when the type certificate is issued and the price is at or below 75% of the updated base value. Re-run `python analysis.py` with each 10-Q.